

Speed and velocity

The time something takes to move a distance is called speed. Velocity is speed in a particular direction. A rocket's velocity is much greater than a snail's!

A snail moves at about 0.003 miles (0.005 km) per hour.



A fast sprinter runs at about 27 miles (43 km) per hour.



A race car reaches speeds of up to 257 miles (413 km) per hour.



A rocket needs to travel at 25,000 miles (40,000 km) per hour.



Down to earth

When you jump, a strong force called gravity pulls you back down to Earth. Earth's gravity gets weaker as you travel away from it. On the Moon, gravity is much less strong, so astronauts can jump high with heavy packs on their backs before gravity pulls them back down again.



Pushing is a kind of force. The harder the boy pushes, the bigger the force on the go-cart.



More pushing power is needed to start the go-cart than to keep it moving.



Quick stop

Without friction between its wheels and the ground, this truck would slip all over the road. The tires have thick patterns, called tread, to help them grip the ground.



Good performance

At high speeds, air rushes over the top of a racing car, pushing it down onto the track. This makes the wide tires grip the track better, so the car can take bends faster than an ordinary car.

